

RESEARCH ARTICLE

Real-Time Object Detection Using Low-Resolution Thermal Camera for Smart Ventilation Systems

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ABSTRACT Deep learning-based object detection research has primarily evolved around RGB camera imagery. While state-of-the-art models like YOLO, SSD, and RetinaNet demonstrate high accuracy in RGB images, their direct application to thermal imagery faces performance degradation due to the inherent differences between RGB and thermal characteristics, coupled with computational demands unsuitable for embedded environments. These challenges are particularly pronounced when using low-resolution thermal sensors, where object boundary ambiguity and thermal pattern uncertainty become significant obstacles. We propose FLARE (Fast and Lightweight Architecture for Real-time Estimation), an ultra-lightweight deep learning model for real-time object detection in low-resolution (32×24) thermal images. FLARE implements a Feature Compression Block for efficient feature map compression and integration, along with a Spatial Denoise Block that enables adaptive processing across temperature regions to handle thermal sensor noise characteristics. To address the scarcity of thermal data, we developed the TOI (Thermal Object Insertion) data augmentation technique, which generates new training data while preserving thermal pattern characteristics. A composite loss function combining box, classification, and confidence losses was implemented to enhance object detection performance. Implementation on an STM32 MCU demonstrated that our proposed model achieved a higher mAP compared to existing lightweight models, while reducing Flash usage by 40.39 %, RAM by 49.02 %, inference time by 35.18 %, and MAC operations by 38.00 %. Comparative experiments with YOLOv8n, a state-of-the-art lightweight object detection model, on Raspberry Pi 4B showed that FLARE achieved higher mAP (0.72% improvement) while utilizing only 4.72% of YOLOv8n's memory consumption and achieving 11.34 times faster execution speed.

INDEX TERMS Computer vision, deep learning, embedded systems, object detection, thermal imaging, data augmentation.

I. INTRODUCTION

In computer vision, deep learning-based object detection technologies such as YOLO [1], SSD [2], and Faster R-CNN [3] have achieved remarkable progress with real-time, high-accuracy detection in RGB camera images. However, this advancement has primarily focused on RGB imagery, with limited research on specialized imaging sensors like thermal cameras. While infrared (IR) images are particularly useful in low-light or harsh environments, they typically suffer from low resolution, high noise levels, and small object sizes.

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Consequently, there is a need for research and development of object detection models specifically tailored for infrared image data.

IR sensor-based object detection technology is widely utilized across various fields, including CCTV [4], the defense industry [5], and industrial equipment. While these IR sensors offer the advantage of operability in low-light or harsh environments, their standalone use is limited due to low resolution and high noise levels. Overcoming these limitations requires the development of optimized object detection models tailored to the characteristics of each application field.

We aim to study object recognition methods for effective application in smart ventilation systems that automatically

adjust ventilation levels based on indoor occupancy by incorporating IR sensors into home ventilation systems.

The COVID-19 pandemic has heightened consumer awareness of indoor air quality, increasing demand for smart ventilation systems that automatically adjust based on indoor occupancy levels.

Thermal sensor-based occupancy detection offers a privacy-preserving solution, operating reliably regardless of lighting conditions. However, the high cost of high-resolution thermal sensors necessitates the use of low-resolution sensors to maintain product price competitiveness. This presents two major challenges: First, thermal images only represent temperature distribution, making object boundaries ambiguous and lacking texture information, which complicates the direct application of existing detection models. Second, the limited computing power and memory resources in industrial settings pose difficulties for real-time operation of complex deep learning models.

To address these challenges, this paper proposes FLARE, a novel lightweight object detection model optimized for embedded environments. The proposed model incorporates a Feature Compression Block for efficient feature map compression and integration, and a Spatial Denoise Block enabling adaptive processing across temperature regions while considering thermal sensor noise characteristics. Additionally, we developed the TOI (Thermal Object Insertion) data augmentation technique to address the scarcity of thermal data, and implemented parameter quantization and computation optimization techniques for efficient real-time processing in embedded environments.

II. RELATED WORK

A. TRADITIONAL THERMAL OBJECT DETECTION

Traditional thermal object detection methods utilize image processing techniques such as thresholding [6], edge detection [7], and region segmentation [8] to identify objects based on temperature differences. These methods employ fixed threshold values or predefined rules to separate objects from the background. While effective in controlled environments, they show significant limitations when applied to low-resolution thermal images where object boundaries are ambiguous and temperature patterns are complex. Furthermore, these approaches often lack robustness against environmental variations and sensor noise, making them unsuitable for practical applications.

B. DEEP LEARNING-BASED OBJECT DETECTION

The advancement of deep learning has led to the development of two main approaches in object detection: two-stage and single-stage detectors. Two-stage detectors such as R-CNN [9], Fast R-CNN [10], Faster R-CNN, Mask R-CNN [11] first generate region proposals followed by classification and bounding box regression.

While these models achieve high accuracy, their computational complexity makes them impractical for real-time

embedded applications. Single-stage detectors like YOLO, SSD, and RetinaNet [12] provide faster inference speeds by performing object localization and classification simultaneously. The latest version, YOLOv8 [13], has demonstrated significant improvements in both accuracy and computational efficiency. However, these models still require substantial computational resources when applied to thermal images.

C. LOW-RESOLUTION THERMAL OBJECT DETECTION

Recent studies have explored various approaches to address object detection challenges in low-resolution thermal imaging. Vandersteegen et al. proposed “Person Detection Using an Ultra Low-resolution Thermal Imager on a Low-cost MCU” [14]. Their study achieved 91.62 % accuracy with 46ms inference time on an STM32 microcontroller platform. However, their approach showed limited adaptability to various thermal conditions due to its fixed feature extraction methodology.

Zhang et al. introduced “Detecting Small Objects in Thermal Images Using Single-Shot Detector” [15], proposing DDSSD (Dilation and Deconvolution Single Shot Multibox Detector). While this study improved small object detection performance, the increased computational complexity from expanded convolution and deconvolution operations hindered real-time processing capabilities.

Recently, Tu et al. presented “Transformer-Based YOLOv5 for Small Object Detection in UAV Thermal Imagery” [16]. Although the transformer architecture enhanced small object detection performance, the high computational requirements limit its applicability in embedded systems.

D. MODEL COMPRESSION AND OPTIMIZATION

Mobile-optimized architectures like MobileNetV2 [17] have introduced efficient components such as depthwise separable convolutions and inverted residual structures. While these innovations reduce computational complexity, they struggle to handle the unique characteristics of thermal imaging, such as temperature gradients and sensor noise. Particularly in low-resolution thermal images, existing lightweight models suffer significant performance degradation due to ambiguous object boundaries and complex thermal patterns.

However, existing studies have not yet achieved an optimal balance between accuracy and computational efficiency in low-resolution thermal object detection. While model compression is essential for real-time processing in embedded environments, simply applying conventional lightweight techniques or compression methods proves insufficient for effectively handling the unique characteristics of thermal images. Overcoming these limitations requires novel network architectures and optimization strategies specifically designed for low-resolution thermal imaging characteristics.

III. PROPOSED METHOD

As shown in Figure 1, the proposed system automatically adjusts ventilation rates based on occupant count detected

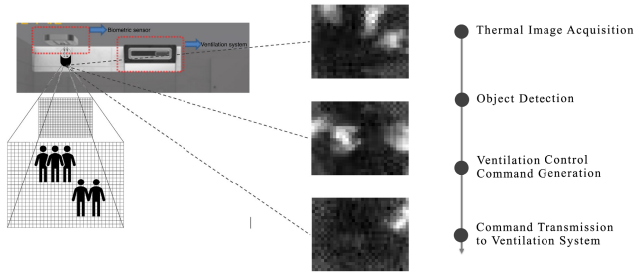


FIGURE 1. Thermal sensor-based autonomous ventilation system.

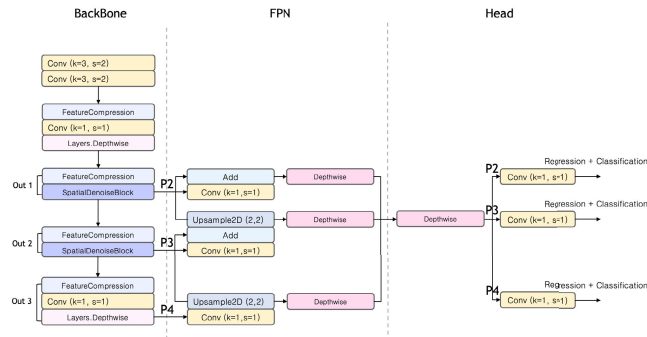


FIGURE 2. Overall structure of the proposed object detection model.

through thermal sensors. Low-resolution thermal images captured indoors serve as input to the object detection model, while detected occupancy data is transmitted to state prediction and ventilation control modules to generate optimal control commands. These commands are executed through networked ventilation units that share control information to maintain integrated management of indoor ventilation status. The system emphasizes low cost, low power consumption, and miniaturization for deployment across various indoor environments including homes, offices, and schools.

A. MODEL ARCHITECTURE

As shown in Figure 2, the proposed FLARE model consists of a backbone network, feature pyramid network, and detection head. Each component is optimized for efficient object detection in low-resolution thermal images and designed for real-time processing in MCU environments.

As shown in Figure 3, the backbone network incorporates Feature Compression Block and Spatial Denoise Block to extract multi-scale feature maps from input images. The Spatial Denoise Block processes input feature maps through two paths: Part A and Part B. Part A functions as a skip connection path, preserving inherent features while transmitting to subsequent computational stages, while Part B performs noise removal and feature enhancement.

1) BACKBONE NETWORK

The Feature Compression Block enhances feature extraction efficiency through a structure combining 1×1 convolution kernels with stride 2. Compared to conventional CNN configurations using 3×3 or 5×5 kernels with stride 1, the proposed block structure simultaneously performs channel-wise

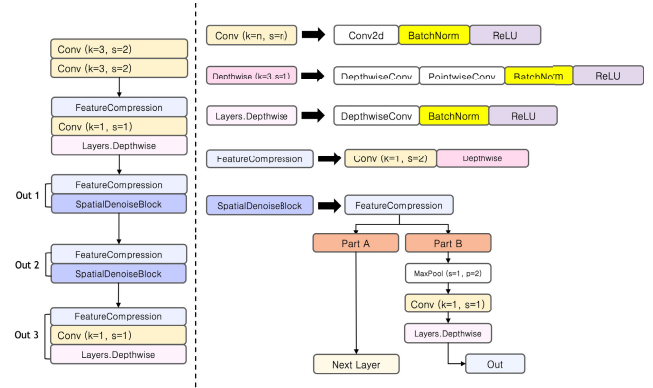


FIGURE 3. Structure of the proposed backbone network.

information integration and spatial feature compression while effectively expanding the receptive field.

2) FEATURE PYRAMID NETWORK

Based on the structure proposed at CVPR 2017, the feature pyramid network [18] combines feature maps extracted from the backbone network in a bottom-up manner, as shown in Figure 2. This process facilitates the detection of small objects and objects with ambiguous boundaries in low-resolution thermal images.

3) DETECTION HEAD

The detection head processes pyramid feature maps to simultaneously perform object localization and classification. It conducts anchor-based detection at each spatial position, predicting bounding box coordinate offsets, class-wise probabilities, and objectness scores. Multi-scale predictions are integrated into a single tensor for final output, structured to include batch size, number of anchor boxes, and prediction parameters (4 coordinates, number of classes, objectness score).

Existing deep learning models face challenges in MCU environments including excessive computational requirements, high memory usage, poor cache efficiency, and inadequate thermal image processing capabilities. While YOLO models demand substantial computational resources and MobileNet introduced lightweight architectures, they remain suboptimal for MCU environments. In contrast, the proposed model enables real-time operation on low-specification MCUs through its lightweight structure and optimized computations, with specialized architecture for efficient thermal data processing.

B. ANCHOR BOX DESIGN

The anchor box design of FLARE is optimized for low-resolution thermal image characteristics and MCU environment efficiency. Unlike existing models such as YOLO and SSD that utilize wide-ranging scales and aspect ratios, this paper restricts the range to specific dimensions (32×32 – 72×72) to enhance memory efficiency.

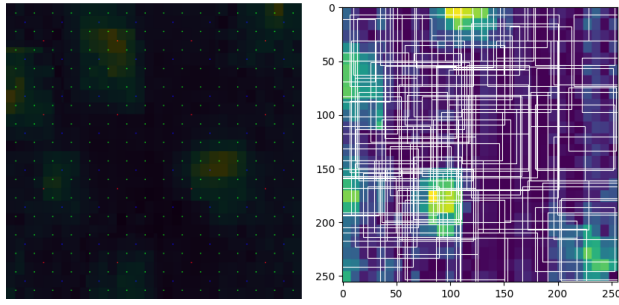


FIGURE 4. Anchor box distribution across stride levels (16, 32, 64) with sample thermal images.

2 As shown in Figure 4, anchor boxes are generated at three stride levels (16, 32, 64) with the following specifications:

- *Stride 16:* Size range 32×32 to 52×52 with aspect ratios 0.75, 1.25
- *Stride 32:* Size range 42×42 to 72×72 with aspect ratios 0.75, 1.25
- *Stride 64:* Fixed size 72×72 with aspect ratios 0.5, 1.0, 1.5

Figure 4 demonstrates the distribution of anchor boxes across stride levels with 100 randomly selected examples. This hierarchical design effectively captures object size and pose variations in thermal images.

Anchor box matching employs a three-stage strategy using IoU thresholds of 0.5 and 0.45. Matches are considered positive when $\text{IoU} \geq 0.5$, negative when $\text{IoU} < 0.45$, and ignored during training for values in between. Box encoding uses normalized relative offsets for center coordinates and dimensions with variance values [0.1, 0.1, 0.2, 0.2].

C. LOSS FUNCTION DESIGN

The proposed model employs a composite loss function tailored for low-resolution thermal object detection. The loss function is designed with three main components to effectively learn object location, class, and presence.

1) BOX LOSS

For bounding box coordinate prediction, we adopt the Huber loss [19] which is robust against outliers:

$$L_{\text{box}} = \begin{cases} \frac{1}{2} (y_{\text{true}} - y_{\text{pred}})^2, & \text{if } |y_{\text{true}} - y_{\text{pred}}| \leq \sigma \\ \sigma \left(|y_{\text{true}} - y_{\text{pred}}| - \frac{1}{2}\sigma \right), & \text{otherwise} \end{cases} \quad (1)$$

Here, y represents the ground truth box coordinates, $\hat{y}$ represents the predicted box coordinates, and δ is a hyperparameter controlling the transition between MSE and MAE. This loss is calculated only for regions where objects are present (positive samples).

2) CLASSIFICATION LOSS

To address class imbalance, we employ Focal Loss:

$$L_{\text{cls}} = -\alpha_t (1 - p_t)^\gamma \log(p_t) \quad (2)$$

Here, p represents the predicted probability for the actual class, α is the weighting factor for class imbalance, and γ is the focusing parameter that controls loss amplification for difficult samples.

3) CONFIDENCE LOSS

The confidence loss for object presence prediction is defined as:

$$L_{\text{conf}} = -\gamma \alpha \log(x) - (1 - \gamma) \beta \log(1 - x) \quad (3)$$

Here, α and β represent weights for positive and negative samples respectively, using binary cross-entropy with predicted probability x and ground truth label $y \in \{0, 1\}$.

4) FINAL TOTAL LOSS

The final loss function combines these three components and is defined as:

$$L_{\text{total}} = L_{\text{box},\text{pos}} + L_{\text{cls},\text{pos}} + L_{\text{cls},\text{neg}} + L_{\text{conf},\text{pos}} + L_{\text{conf},\text{neg}} \quad (4)$$

Here, $L_{\text{cls},\text{pos}}$ and $L_{\text{cls},\text{neg}}$ represent classification losses for positive and negative samples, while, $L_{\text{conf},\text{pos}}$ and $L_{\text{conf},\text{neg}}$ represent confidence losses for positive and negative samples.

D. TOI DATA AUGMENTATION

To address data imbalance in object detection model training, which is particularly pronounced in thermal datasets containing numerous empty images or images with small objects, we propose TOI (Thermal Object Insertion), a data augmentation technique for low-resolution thermal object detection. TOI is designed to reflect the unique characteristics of thermal images: continuity in thermal energy distribution and ambiguity in object boundaries.

1) ALGORITHM OVERVIEW

Figure 5 shows the overall structure of the proposed TOI algorithm, which generates new training samples using input dataset images, bounding box coordinates, class labels, and object count information. This algorithm focuses not only on increasing the quantity of training data but also on improving data quality by preserving essential thermal characteristics in thermal images. In particular, this augmentation technique is applied to the FLARE model and includes the following key technical features:

- 1) **IoU Threshold-based Placement:** The algorithm introduces an IoU threshold τ , allowing object placement only when the IoU with existing objects is below τ . This prevents unnecessary or excessive object overlap and maintains object distributions similar to real situations. Specifically, it controls the placement so that candidate objects are sufficiently separated from existing objects, enabling realistic spatial arrangements.
- 2) **Dynamic Temperature-aware Padding:** The TOI algorithm calculates dynamic padding sizes considering object dimensions and surrounding background

Algorithm 1 TOI with Thermal Augmentation for FLARE**Require:**

- Dataset $\mathcal{D} = \{(x_i, y_i, T_i)\}_{i=1}^N$, where
- x_i : input image (supports both 1-channel and 3-channel),
 - T_i : corresponding thermal map,
 - $y_i = \{(b_j, c_j)\}_{j=1}^{M_i}$, where b_j is a bounding box and c_j is the class label.

Hyperparameters:

- Maximum objects M , IoU threshold τ (e.g., $\tau \in [0.01, 0.08]$, adjustable),
- Padding function $P(\cdot)$,
- Blur/normalization function $G(\cdot)$,
- Horizontal flip probability p_h (e.g., 0.5),
- Vertical flip probability p_v (e.g., 0.5),
- Scaling probability p_s (e.g., 0.4),
- Brightness adjustment range (e.g., add rand(-10, 5)),
- Contrast adjustment range (e.g., multiply by rand(0.8, 1.2)),
- Scaling factor range (e.g., sample $s \leftarrow \text{rand}(0.8, 1.1)$).

Note: The above example values are for illustration only. They should be set according to user requirements or dataset characteristics.

Ensure: Augmented dataset $\mathcal{D}'$

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1:  $\mathcal{D}' \leftarrow \emptyset$ 
2: for each sample  $(x, y, T)$  in  $\mathcal{D}$  do
3:   Initialize augmented image  $x' \leftarrow x$ , thermal map  $T' \leftarrow T$ , and object set  $y' \leftarrow y$ 
4:   Let  $M' \leftarrow |y|$ 
5:   while  $M' < M$  do
6:     Donor Selection: Sample a donor sample  $(x_d, y_d, T_d)$  from  $\mathcal{D}$  with  $|y_d| > 0$ 
7:     Randomly select an object  $(b_d, c_d)$  from  $y_d$ 
8:     Apply a small random jitter to  $b_d$  to obtain  $b'_d$ 
9:     Compute IoU values between  $b'_d$  and each bounding box in  $y'$ 
10:    if  $\max_{c \in y'} \{\text{IoU}(b, b'_d)\} < \tau$  then ▷ IoU Threshold-based Placement
11:      Patch Determination: Compute padding size  $p \leftarrow P(b'_d, T_d)$ 
12:      Expand bounding box:  $b_d \leftarrow \text{expand}(b'_d, p)$ 
13:      Extract image patch:  $x_{\text{patch}} \leftarrow x_d[b_d]$ 
14:      Extract thermal patch:  $T_{\text{patch}} \leftarrow T_d[b_d]$ 
15:      Spatial Transformation:
16:        Brightness Adjustment:  $x_{\text{patch}} \leftarrow x_{\text{patch}} + \text{rand\_brightness}$ 
17:        Contrast Adjustment:  $x_{\text{patch}} \leftarrow x_{\text{patch}} \times \text{rand\_contrast}$ 
18:        Horizontal Flip (with probability  $p_h$ ):
19:        Vertical Flip (with probability  $p_v$ ):
20:        Scaling (with probability  $p_s$ ):
21:        Sample scale factor  $s \leftarrow \text{rand\_scale}$ .
22:        Resize  $x_{\text{patch}}$  by  $s$  and update  $b'_d$  accordingly.
23:        Thermal Pattern Preservation:
24:        Adapt thermal patch:  $T_{\text{patch}} \leftarrow G(T_{\text{patch}}, \text{intensity}(b'_d, T_d))$ 
25:        Normalize thermal patch:  $T_{\text{patch}} \leftarrow \text{Normalize}(T_{\text{patch}}, \mu_{bg}(T'))$ 
26:        Insertion:
27:        Replace region  $b_d$  in  $x'$  with  $x_{\text{patch}}$ .
28:        Replace corresponding region in  $T'$  with  $T_{\text{patch}}$ .
29:        Update object set:  $y' \leftarrow y' \cup \{(b_d, c_d)\}$  and set  $M' \leftarrow M' + 1$ .
30:      end if
31:    end while
32:   $\mathcal{D}' \leftarrow \mathcal{D}' \cup \{(x', y', T')\}$ 
33: end for
34: return  $\mathcal{D}'$ 

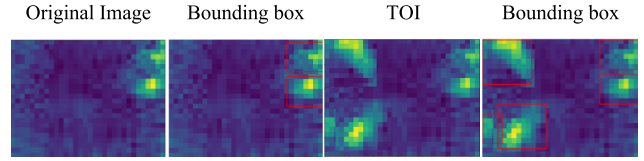
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FIGURE 5. TOI data augmentation algorithm.

thermal distribution to maintain thermal distribution continuity around objects. By using padding function $P(\cdot)$ to expand object regions in donor images, it ensures inserted objects blend naturally with the original thermal image background. This process guarantees smooth temperature transitions, contributing to the generation of augmented data similar to real thermal images.

- 3) **Thermal Pattern Preservation:** To maintain natural patterns in thermal images, the algorithm applies adaptive Gaussian blur and background temperature normalization techniques. The blur kernel size is dynamically adjusted based on object thermal intensity, and inserted object temperature values are normalized to match the background mean temperature $\mu_{bg}(T')$. This enables natural thermal energy transitions between objects and backgrounds, resulting in thermal patterns similar to real thermal images.

Figure 6 illustrates a comparative analysis of images before and after the application of the Thermal Object Insertion (TOI) algorithm. Traditional image augmentation techniques, such as simple transformations like rotation and flipping, or object insertion through cut-and-paste [20], CutMix [21],

**FIGURE 6.** Comparison before and after applying TOI technique.

and MixUp [22], have been widely used for generating augmented datasets. However, these methods fail to account for the unique characteristics of thermal images, leading to several critical issues.

First, these conventional techniques disrupt the natural thermal distribution, causing temperature discontinuity in the augmented images. Second, unnatural object boundaries emerge due to abrupt pixel transitions, as traditional methods do not consider the heat gradient properties of thermal imagery. Third, unrealistic object placement often occurs, resulting in spatial inconsistencies that rarely appear in real-world thermal imaging conditions.

Unlike conventional methods, TOI employs IoU-based placement to regulate object positioning, ensuring minimal overlap with existing structures and reducing unrealistic spatial distortions. Additionally, temperature-aware padding dynamically adjusts to maintain smooth thermal distributions, while thermal pattern preservation prevents abrupt temperature discontinuities, effectively blending inserted objects into the scene.

Additionally, dynamic temperature-aware padding preserves the continuity of thermal distributions by adjusting padding size based on object dimensions and surrounding temperature gradients. Lastly, thermal pattern preservation mechanisms utilize adaptive Gaussian blur and background temperature normalization to smooth temperature transitions, ensuring that inserted objects blend seamlessly into the scene.

As demonstrated in Figure 6, a direct comparison between the original image (left) and the TOI-processed image (right) highlights the effectiveness of TOI in generating realistic thermal augmentations. The inserted objects blend harmoniously with the background, and the temperature distribution around objects exhibits natural patterns, closely resembling real thermal images. These enhancements contribute significantly to improving the reliability and robustness of object detection models in thermal imaging applications.

Figure 7 demonstrates various examples of augmented data generated through the TOI algorithm. While existing simple image transformation techniques fail to address data imbalance issues and offer limited transformations, TOI provides several advantages:

- **Realistic Spatial Arrangement:**

Implements natural object placement similar to real environments through IoU threshold-based positioning.

- **Thermal Distribution Continuity:**

Applies dynamic temperature-aware padding to ensure inserted objects smoothly connect with the original

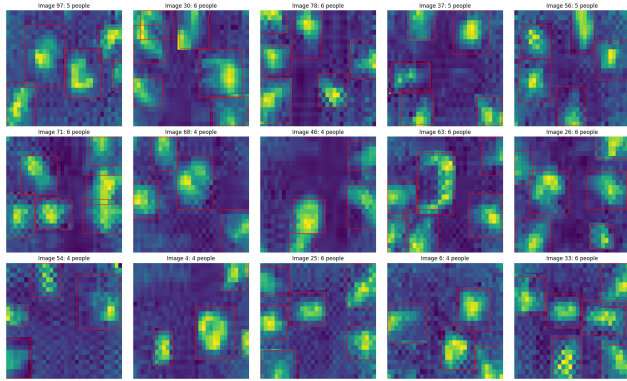


FIGURE 7. Examples of augmented data generated through TOI.

thermal image background, providing results similar to actual thermal camera footage.

- **Data Imbalance Resolution:**

Enhances training data diversity and effectively addresses data imbalance issues by generating various scenario data.

- **Model Robustness Enhancement:**

Improves model generalization performance and strengthens resilience against outliers by combining high-quality augmented data with the original dataset.

Thus, the proposed TOI algorithm, applied as an augmentation technique in the FLARE model, helps achieve high object detection performance by generating realistic object distributions while preserving thermal image characteristics.

IV. EXPERIMENTAL RESULTS

A. EXPERIMENTAL SETUP

1) IMPLEMENTATION ENVIRONMENT

The experimental implementation utilizes an STM32F767ZIT6 MCU [23] based on the ARM Cortex-M7 core. This MCU operates at a frequency of 216 MHz and features 2 MB Flash memory, 512 KB SRAM, along with 16KB Data cache and 16 KB Instruction cache. Image acquisition is performed through an MLX90640 [24] thermal sensor providing a resolution of 32×24 (768 pixels), which communicates with the MCU via an I2C interface.

2) DATASET CONSTRUCTION

The dataset is collected using a ceiling-mounted downward-facing configuration to reflect the actual operating environment of smart ventilation systems. The data collection environment is designed to maintain ceiling heights of 2-3 m and indoor temperatures ranging from 18-32 °C, incorporating various occupancy patterns, ventilation states, and lighting changes. Additionally, it includes challenging scenarios such as complex thermal characteristics, partial occlusion, and various thermal contrast conditions to evaluate model robustness.

As shown in Figure 8, the dataset consists of low-resolution thermal image data with a resolution of 32×24 , presenting challenges in object recognition. The low resolution causes

severe pixelation, resulting in unclear object boundaries and reduced overall thermal contrast, leading to degraded detection performance. Furthermore, sensor noise makes it difficult to distinguish objects from the background, creating additional challenges in detecting small objects or those with low emissivity. Addressing these low-resolution and noise issues effectively requires a substantial amount of diverse training data.

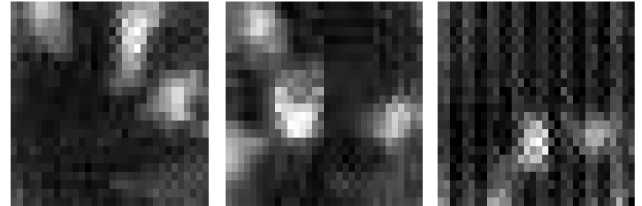


FIGURE 8. Example of experimental data.

To this end, we have collected a total of 10,000 images and divided them into training data (8,000 images, 80 %), validation data (1,000 images, 10 %), and test data (1,000 images, 10 %). Additionally, by utilizing the TOI data augmentation technique to expand the training dataset to 50,000 images, we have secured more diverse training data to address the challenging issues of low resolution and noise mentioned above.

3) TRAINING DETAILS

The loss function combines three components. For bounding box coordinate prediction, we employed Huber Loss ($\delta = 2.0$) for robustness against outliers. To address class imbalance issues, we implemented Focal Loss ($\alpha = 0.25$, $\gamma = 2.0$). The confidence loss for object existence prediction was based on Binary Cross Entropy with positive and negative sample weights ($\alpha = 1.3$, $\beta = 1.0$). The model training employed the Adam optimizer with an initial learning rate of 0.0008. We set the batch size to 16 and trained the model for up to 100 epochs, with Early Stopping applied based on validation performance.

4) BENCHMARK MODELS

To evaluate performance, we initially aimed to compare the proposed method with the latest YOLOv8n, known for its excellent real-time performance. However, due to the high hardware complexity of YOLOv8n, it could not be implemented on the resource-constrained STM32F767. Therefore, two separate comparison strategies were adopted. First, the proposed method was compared with MobileNetv2-SSD and YOLOv1, both officially supported and implementable on the STM32F767 via ST's X-CUBE-AI [25]. Then, to compare with YOLOv8n, we additionally evaluated the proposed method on the Raspberry Pi 4B [26] platform.

B. PERFORMANCE ANALYSIS

Table 1 compares the performance of existing methods and the proposed approach in low-spec MCU environments

TABLE 1. Performance comparison on STM32F767ZIT6.

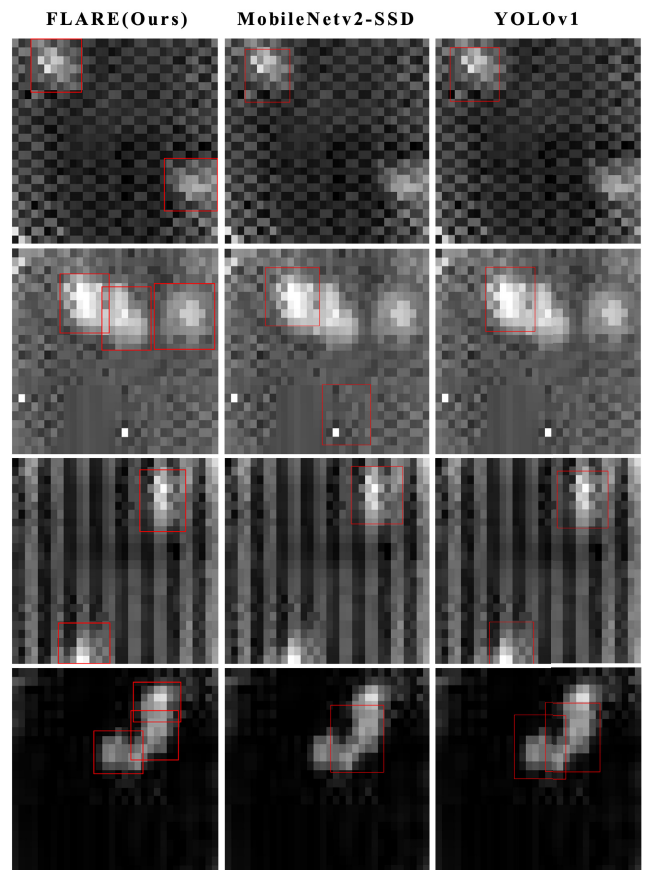
	Ours(FLARE) 1channel	Ours(FLARE) 3channel	MobileNetv2- SSD	YOLOv1
Used Flash	281.30 KiB	282.80 KiB	471.87 KiB	276.73 KiB
Used RAM	256.00 KiB	256.00 KiB	502.13 KiB	449.17 KiB
Input shape	(256, 256, 1)	(256, 256, 3)	(256, 256, 3)	(256, 256, 3)
Total MAC	44,331,902	54,167,934	71,504,135	110,052,238
Inference Time	303.537 ms	345.754 ms	468.276 ms	559.481 ms
mAP@0.50	95.03 %	95.21 %	52.03 %	87.33 %
True Positive	2092	2103	1085	1660
False Positive	83	88	173	103
False Negative	69	58	1076	501

with very limited resources. Performance evaluations against STMicroelectronics officially supported MobileNetv2-SSD and YOLOv1 demonstrate that the proposed FLARE 1-channel model achieves the lowest memory consumption, with Flash usage of 281.30 KiB and RAM usage of 256.00 KiB. In terms of MAC operations, FLARE requires only 44.3 million, corresponding to 62.0 % of MobileNetv2-SSD and 40.3 % of YOLOv1.

In MCU environments, MobileNetv2-SSD and YOLOv1 suffer from low cache efficiency and slow inference speeds due to excessive computations—468.3 ms (71.5M MAC) and 559.5 ms (110.1M MAC), respectively—making them unsuitable for real-time applications. In contrast, FLARE achieves real-time processing with 303.5 ms inference time, while maintaining superior detection accuracy (95 % mAP). This is enabled by its efficient architecture composed of a Feature Compression Block for channel-wise information integration and spatial feature reduction, a Spatial Denoise Block that preserves key input features via skip connections while reducing noise, and a Feature Pyramid Network for multi-scale feature extraction. In addition, when evaluated at a confidence threshold of 0.5, FLARE demonstrated significantly fewer false positives (FP: 83) and false negatives (FN: 69) compared to MobileNetv2-SSD (FP: 173, FN: 1076) and YOLOv1 (FP: 103, FN: 501), while also recording the highest number of true positives (TP: 2092). These results indicate that FLARE provides robust and reliable performance even under constrained edge environments.

Figures 9, 10, and 11 present a comparative analysis of object detection performance across challenging environments. MobileNetv2-SSD exhibits a high false positive rate in complex scenarios involving overlapping objects and thermal interference near air conditioners and windows. In high-noise environments, it struggles to accurately delineate object contours, leading to reduced detection accuracy. Similarly, YOLOv1 frequently fails to detect objects positioned near the frame edges and misclassifies objects in regions with rapid temperature fluctuations, resulting in increased false negatives.

In contrast, FLARE consistently achieves stable detection performance in these challenging conditions, leveraging its

**FIGURE 9.** Experimental detection results under varying noise conditions.

Feature Compression Block and Spatial Denoise Block for enhanced feature extraction and noise suppression. In object overlap scenarios (Figure 11), FLARE effectively distinguishes individual object boundaries, minimizing false positives and improving detection precision. In edge-located object scenarios (Figure 10), it maintains high accuracy, successfully detecting partially visible objects that other models frequently miss. Furthermore, under severe sensor noise conditions (Figure 9), FLARE accurately captures object thermal patterns through robust noise filtering and adaptive feature enhancement, ensuring reliable detection.

These results confirm FLARE's adaptability to real-world operational challenges, demonstrating its robustness in detecting objects under varying environmental conditions, including noise, thermal interference, and spatial occlusion.

Table 2 presents the performance comparison of detection models on Raspberry Pi 4B environment. For comparison with state-of-the-art object detection models, performance evaluation was conducted against YOLOv8n on Raspberry Pi 4B environment. Figure 12 shows the performance comparison results with YOLOv8n. In comparative experiments on Raspberry Pi, while YOLOv8n achieved a mAP of 94.9 %, FLARE achieved a higher mAP of 95.62 % while reducing Flash memory usage to 4.72 % and RAM usage to 0.85 % of YOLOv8n, and improving processing speed by 11.3 times.

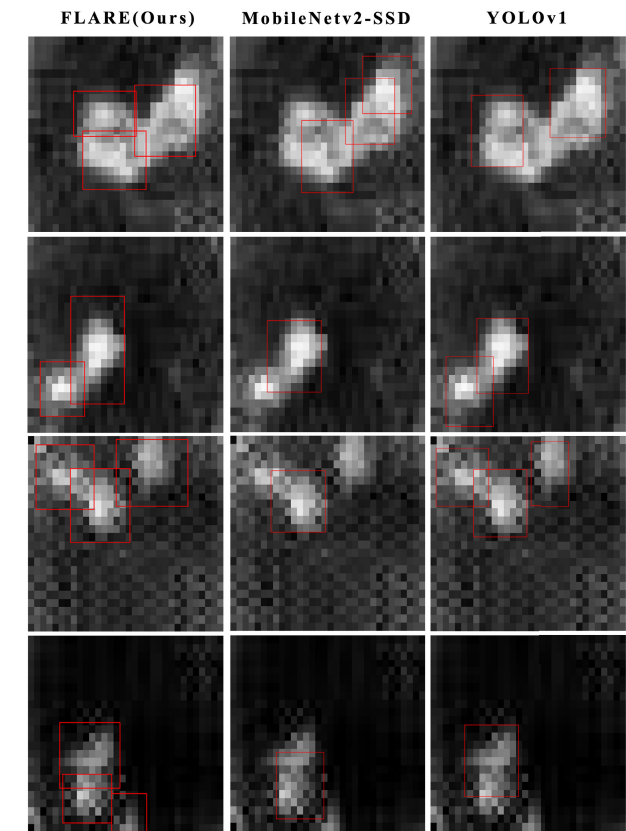


FIGURE 10. Detection results for objects located in peripheral frame regions.

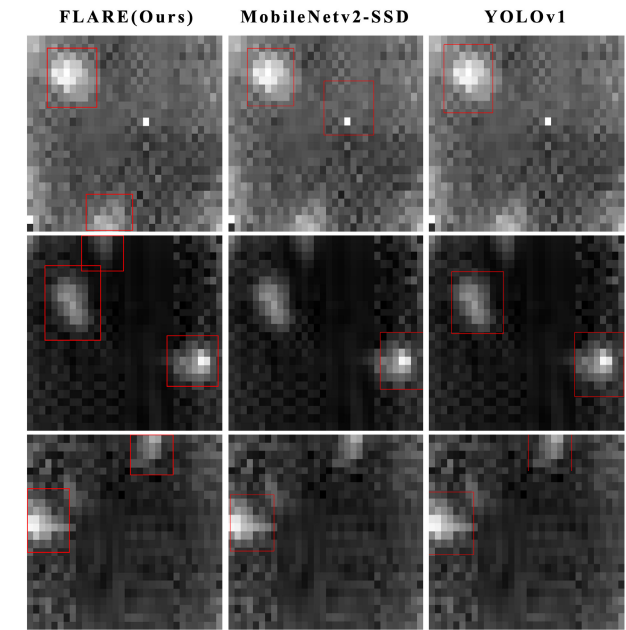


FIGURE 11. Detection performance in scenarios with object occlusion.

In addition, when evaluated at a confidence threshold of 0.5, FLARE recorded fewer false negatives (FN: 58 vs. 265) than YOLOv8n. While YOLOv8n showed fewer false positives (FP: 32), FLARE achieved a higher number of true

TABLE 2. Performance comparison on RaspberryPi 4B 8GB.

	Ours(FLARE)	YOLOv8n
Used Flash	281.65 KiB	5966.25 KiB
Used RAM	280.00 KiB	32940.00 KiB
Input shape	(256, 256, 3)	(256, 256, 3)
Inference Time	10.14 ms	114.99 ms
mAP@0.50	95.62 %	94.90 %
True Positive	2103	1896
False Positive	88	32
False Negative	58	265

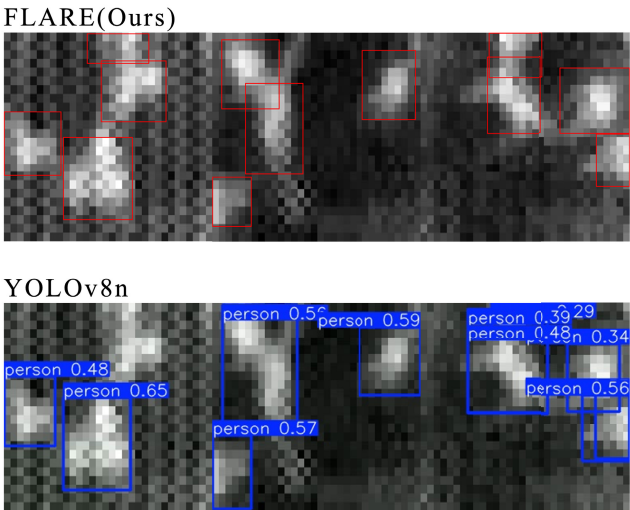


FIGURE 12. Performance evaluation of detection results under diverse environmental conditions.

positives (TP: 2103 vs. 1896), indicating that it correctly detected more target instances.

As shown in Figure 12, FLARE demonstrated stable detection performance even in situations involving object overlap and thermal interference.

V. CONCLUSION

In this paper, we propose FLARE, a lightweight thermal object detection model optimized for low-specification MCU environments. While IR sensor-based object detection technology is widely used across various fields including CCTV, defense industry, and industrial equipment, existing deep learning-based detection methods faced limitations in detecting small objects in low-resolution environments, vulnerability to sensor noise, and real-time processing constraints due to high computational requirements.

To address these issues, we propose the FLARE model. The proposed model includes newly designed Feature Compression Block for channel-wise information integration and spatial feature compression, and Spatial Denoise Block for noise removal and input characteristic preservation, while utilizing Feature Pyramid Network for multi-scale feature extraction.

In experiments on STM32F767ZIT6 MCU, FLARE significantly reduced memory usage (281.30 KiB Flash, 256.00 KiB RAM) and computational requirements (44.3 M MAC) while achieving higher detection accuracy (95.03 % mAP) and faster processing speed (303.5 ms) compared to MobileNetv2-SSD and YOLOv1. Additionally, in comparison with the state-of-the-art YOLOv8n on Raspberry Pi environment, FLARE demonstrated significant improvements by reducing memory usage (21.2 times less Flash, 117.6 times less RAM) and improving processing speed by 11.3 times while achieving higher mAP (95.62 % vs 94.90 %).

Furthermore, through the TOI data augmentation technique, which preserves thermal image characteristics while securing diverse training data, the model demonstrated stable detection performance in various challenging situations such as noise, object overlap, and edge-located objects that may occur in real operational environments. The proposed method significantly improved object detection performance compared to existing methods using low-resolution thermal sensor input on low-specification MCUs, and is expected to be effectively applicable to various thermal object detection applications based on low-specification MCUs, including not only smart ventilation systems but also CCTV, defense industry, and industrial equipment.

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